

```
from google.colab import files

uploaded = files.upload()

image_path = list(uploaded.keys())[0]

print("Image uploaded:", image_path)
```

roi-dimor-B...nsplash.jpg

roi-dimor-BHA91Ia48kY-unsplash.jpg (image/jpeg) - 2956895 bytes, last modified: 9/15/2026 - 100% done
Saving roi-dimor-BHA91Ia48kY-unsplash.jpg to roi-dimor-BHA91Ia48kY-unsplash.jpg
Image uploaded: roi-dimor-BHA91Ia48kY-unsplash.jpg

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread(image_path)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(5,5))
plt.imshow(gray, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis('off')
plt.show()
```

Original Grayscale Image

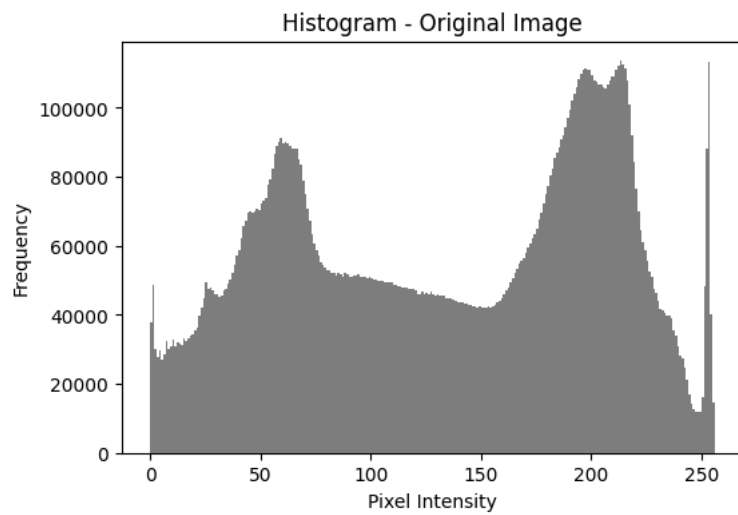


```
plt.figure(figsize=(6,4))

plt.hist(gray.ravel(), bins=256, range=[0,256], color='gray')

plt.title("Histogram - Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.show()
```



```
def contrast_stretch(img):
    min_val = np.min(img)
    max_val = np.max(img)

    stretched = (img - min_val) * (255.0 / (max_val - min_val))

    return stretched.astype(np.uint8)

stretched_img = contrast_stretch(gray)

plt.figure(figsize=(5,5))

plt.imshow(stretched_img, cmap='gray')
plt.title("Contrast Stretched Image")
plt.axis('off')

plt.show()
```

Contrast Stretched Image



```
equalized_img = cv2.equalizeHist(gray)

plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(gray, cmap='gray')
plt.title("Original")
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(equalized_img, cmap='gray')
plt.title("Histogram Equalized")
plt.axis('off')

plt.show()
```

Original



Histogram Equalized



```
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8,8)
)

clahe_img = clahe.apply(gray)

plt.figure(figsize=(5,5))

plt.imshow(clahe_img, cmap='gray')
plt.title("CLAHE Enhanced Image")
plt.axis('off')

plt.show()
```

CLAHE Enhanced Image



```
fig, axs = plt.subplots(1, 3, figsize=(15,4))

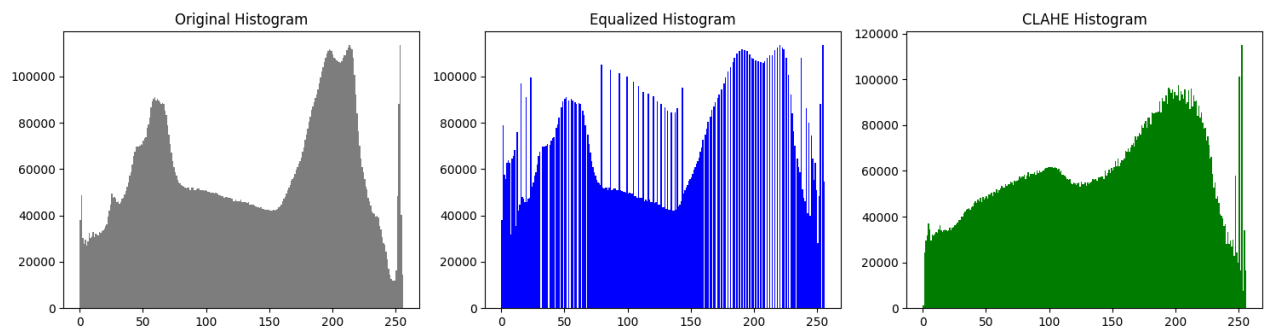
axs[0].hist(
    gray.ravel(),
    bins=256,
    range=[0,256],
    color='gray'
)
axs[0].set_title("Original Histogram")

axs[1].hist(
    equalized_img.ravel(),
    bins=256,
    range=[0,256],
    color='blue'
)
axs[1].set_title("Equalized Histogram")

axs[2].hist(
    clahe_img.ravel(),
    bins=256,
    range=[0,256],
    color='red'
)
axs[2].set_title("CLAHE Enhanced Histogram")
```

```
clahe_img.ravel(),
bins=256,
range=[0,256],
color='green'
)
axs[2].set_title("CLAHE Histogram")

plt.tight_layout()
plt.show()
```



```
fig, axs = plt.subplots(1, 4, figsize=(18,5))

axs[0].imshow(gray, cmap='gray')
axs[0].set_title("Original")
axs[0].axis('off')

axs[1].imshow(stretched_img, cmap='gray')
axs[1].set_title("Contrast Stretched")
axs[1].axis('off')

axs[2].imshow(equalized_img, cmap='gray')
axs[2].set_title("Histogram Equalized")
axs[2].axis('off')

axs[3].imshow(clahe_img, cmap='gray')
axs[3].set_title("CLAHE")
axs[3].axis('off')

plt.tight_layout()
plt.show()
```

